

| Question |     |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks available |     |     |       |       |      |
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|          |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4        | (a) |  | <p><b>Increase in temperature - kinetic theory effects</b><br/>Molecules move randomly<br/>Collisions become more frequent [when heat supplied]<br/>[No change in volume so heat flowing in] causes increase in <math>U</math> / kinetic energy<br/>No work done [as constant volume]<br/><math>T</math> increases with <math>U</math> as temperature proportional to <math>U</math>, or equivalent</p> <p><b>Newton's laws of motion</b><br/>Momentum of molecules increase<br/>Force on molecules = rate of change of momentum (during collision with wall)<br/>Force on wall is equal and opposite to force on molecules<br/>Greater forces during the collisions</p> <p><b>Increase in pressure</b><br/>Molecules collide with walls exerting force on walls and / or pressure<br/>Pressure increases with temperature<br/>Pressure = force on walls per unit area<br/>Mean pressure due to many collisions [and many molecules]</p> <p><b>5-6 marks</b><br/>Comprehensive account including reference to increase in temperature, Newton's laws of motion and increase in pressure.<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i></p> <p><b>3-4 marks</b><br/>Comprehensive account including reference to 2 out of 3 of increase in temperature, Newton's laws of motion and increase in pressure or brief account of all 3 areas.<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i></p> | 6               |     |     | 6     |       |      |

| Question |     |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks available |     |     |       | Maths | Prac |
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|          |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2 | AO3 | Total |       |      |
|          |     |      | <p><b>1-2 marks</b><br/>Comprehensive account including reference to one of increase in temperature, Newton's laws of motion and increase in pressure or limited account of 2 areas.<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i></p> <p><b>0 marks</b><br/><i>No attempt made or no response worthy of credit.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 |     |     |       |       |      |
|          | (b) | (i)  | $\text{rms} = \sqrt{\frac{400^2 + 425^2 + 450^2 + 550^2 + 625^2}{5}} \quad (1)$ $= 497 \text{ [m s}^{-1}\text{]} \quad (1)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 | 2   |     | 2     | 2     |      |
|          |     | (ii) | <p>The expected rms is explained by:</p> $PV = nRT \quad \text{so} \quad P = \frac{nRT}{V}$ $\rho = \text{mass}/V = \frac{n(M_r \times 10^{-3})}{V} \quad \text{use of both equations by substitution (1)}$ <p>substitute these into: <math>P = \frac{1}{3}\rho \overline{c^2}</math>; <math>\sqrt{\overline{c^2}} = \sqrt{\frac{3P}{\rho}}</math>;</p> $\sqrt{\overline{c^2}} = \sqrt{3 \frac{nRT}{V} \frac{V}{n(M_r \times 10^{-3})}} = \sqrt{\frac{3RT}{(M_r \times 10^{-3})}}$ $\sqrt{\overline{c^2}} = \sqrt{\frac{3(8.31)(293)}{(32 \times 10^{-3})}} = 478 \text{ [m s}^{-1}\text{]} \quad (1)$ <p>Valid conclusion with data i.e. yes, the rms speed of the five molecules is slightly higher [about 4% above the expected rms of the gas] (1) <b>ecf</b></p> <p><b>Alternative solution:</b> Use <math>m\overline{c^2} = 3kT</math></p> $\sqrt{\overline{c^2}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23})(293)}{(32)(1.66 \times 10^{-27})}}$ <p>with sensible substitution (1)= 478 [m s<sup>-1</sup>] (1)</p> |                 | 1   | 1   | 3     | 3     |      |

| Question |  |       | Marking details                                                                                                                                                                                                                                                                                                                             | Marks available |          |          |           |          |          |
|----------|--|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |  |       |                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |  | (iii) | Density does not change <b>and</b><br>Pressure increases from the original pressure $p$ to $1.2p$ (1) i.e. an increase of 20%.<br>so<br>$\left(\sqrt{c^2}\right)_{\text{new}} = \sqrt{\frac{3(1.2p)}{\rho}}$ $\left(\sqrt{c^2}\right)_{\text{new}} = \sqrt{1.2} \sqrt{\frac{3p}{\rho}} = \sqrt{1.2} (478) = 524 \text{ m s}^{-1} \quad (1)$ |                 | 2        |          | 2         | 2        |          |
|          |  |       | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                     | <b>6</b>        | <b>5</b> | <b>2</b> | <b>13</b> | <b>7</b> | <b>0</b> |